

Cricket Blaze Game Report

Hanzlah Mehmood Ch
Roll Number: 23i-0799
Section: CS-D
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1 Gameplay Screenshots

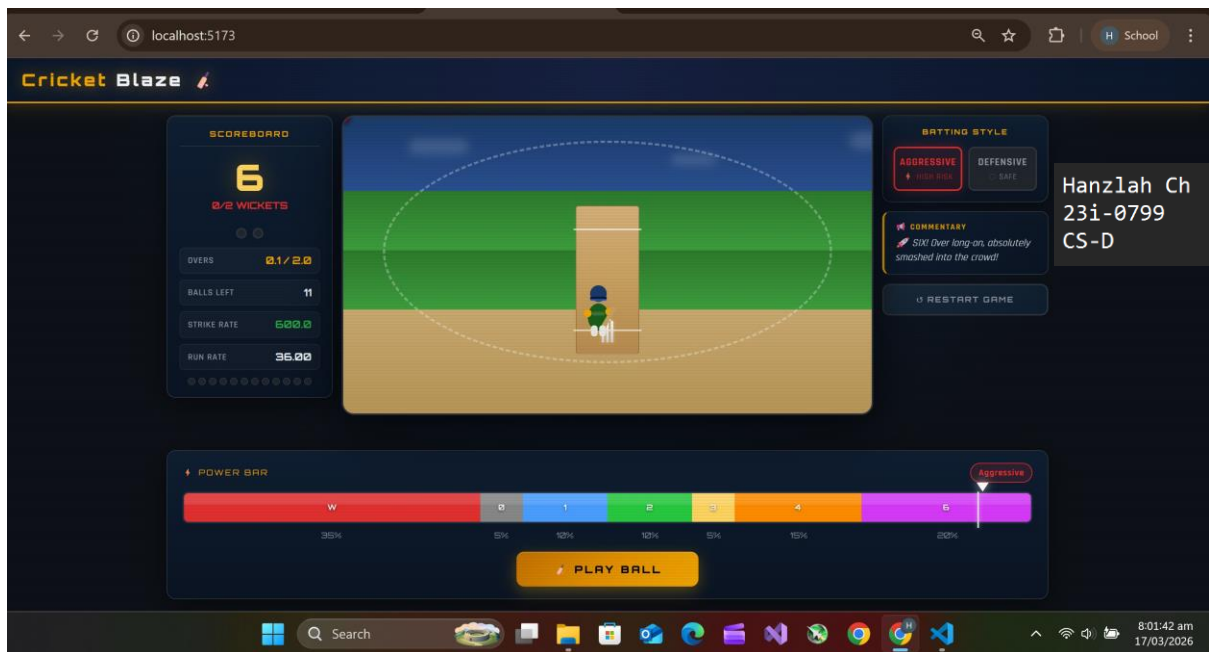


Figure 1: Aggressive batting in action.

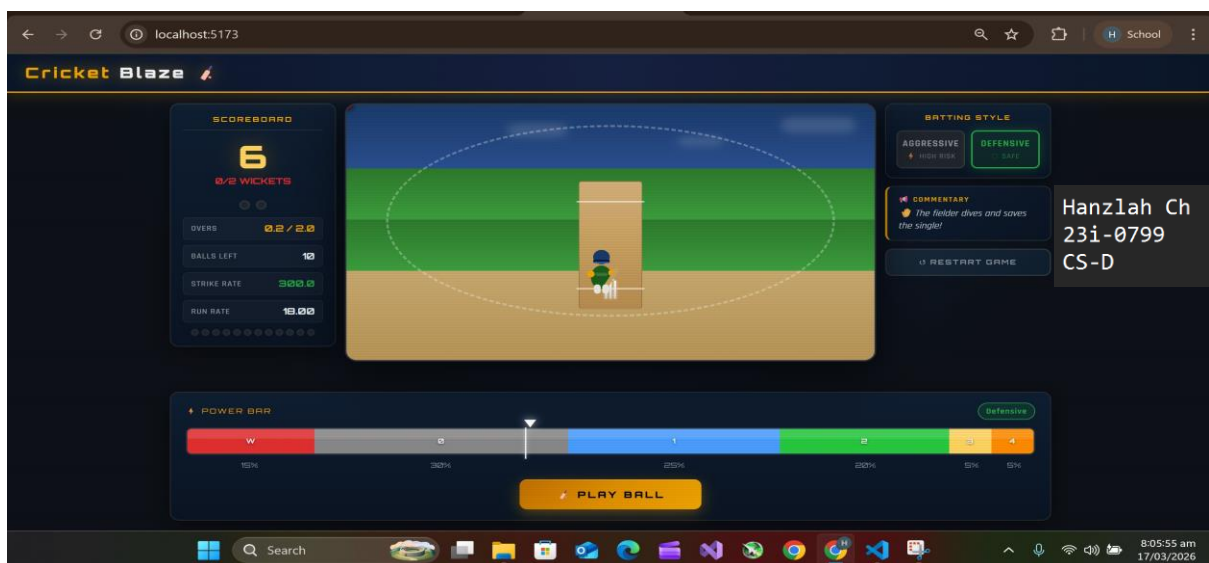


Figure 2: Defensive batting in action.

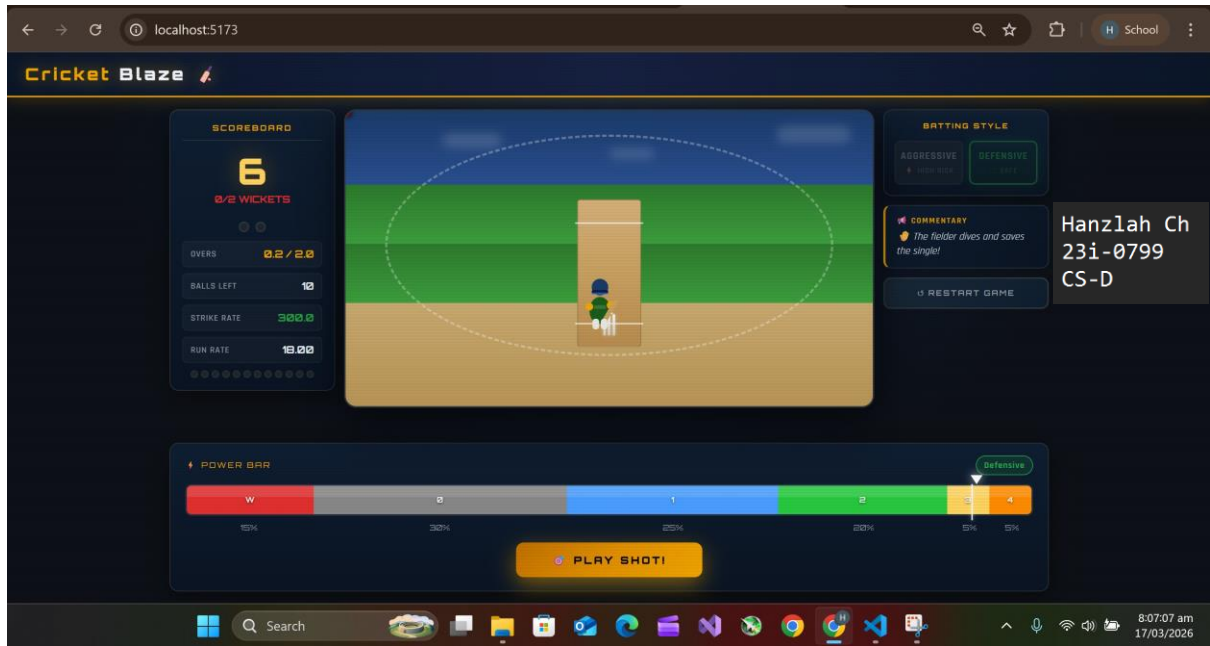


Figure 3: Power bar visible during gameplay.

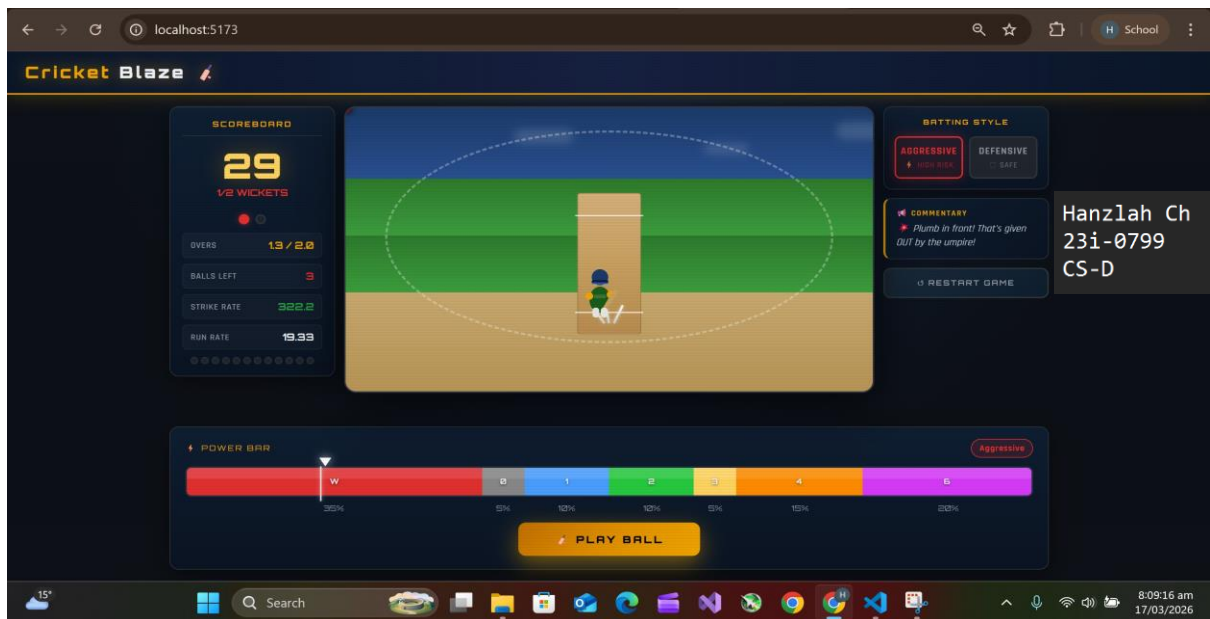


Figure 4: Scoreboard updates reflecting game progress.

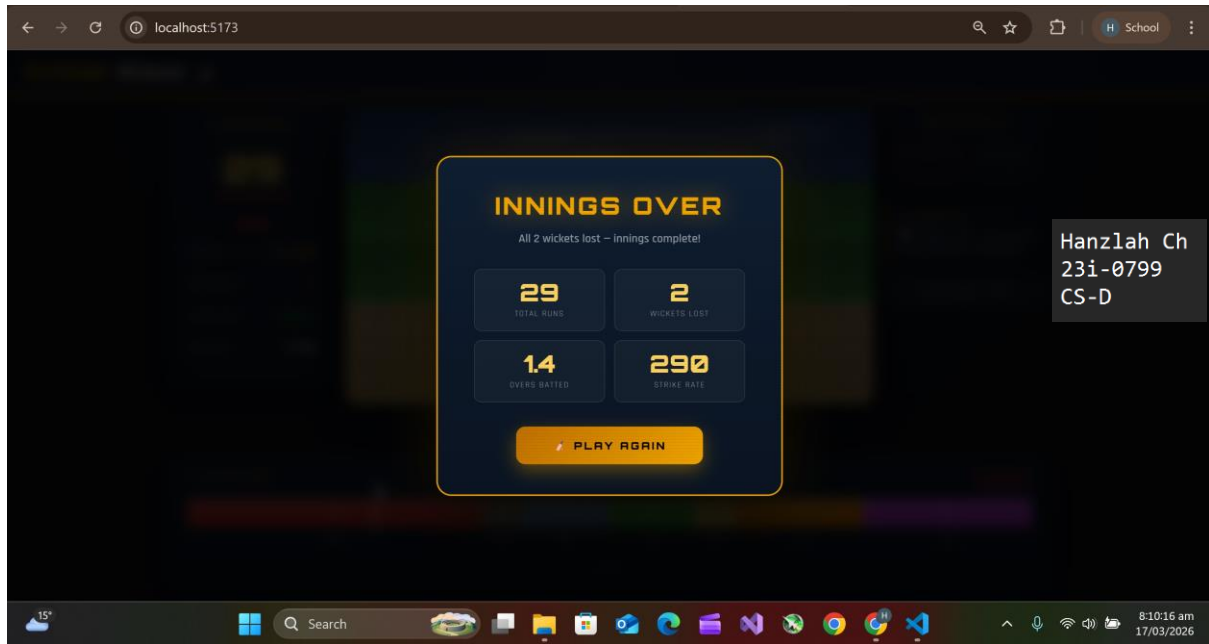


Figure 5: Game over screen.

2 Brief Explanation

2.1 Game Logic and Working Mechanism

The game simulates a short cricket innings with fixed limits on balls and wickets. At each ball, the user selects a batting style and triggers a shot attempt. The slider position on the power bar is captured at shot time and mapped to an outcome (W , 0, 1, 2, 3, 4, or 6 runs). After each outcome, runs/wickets/balls are updated, commentary is displayed, and the game checks for innings completion (all balls used or wickets exhausted).

2.2 Mapping of Probabilities to the Power Bar

Each batting style has its own probability distribution across outcomes:

- **Aggressive:** Higher chance of boundaries and wicket risk.
- **Defensive:** Lower wicket risk, more dots/singles, fewer big shots.

The power bar is divided into segment widths proportional to their probabilities. When the moving slider is stopped, the captured position is compared against cumulative segment boundaries. The segment containing that position determines the final outcome for that delivery.

2.3 Implementation of Animations

Animations are used to make gameplay interactive and readable:

- Slider animation (left-right movement) while waiting for shot timing.
- Ball delivery and post-shot movement animations based on outcome type.
- Batsman swing animation during shot execution.
- Result flash overlays (e.g., OUT, FOUR, SIX) with timed fade/pop effects.
- Stump knock effect for wicket events and smooth UI transitions for controls.

Animation sequencing is controlled through game phases (*idle, bowling, waiting, shot, result*) to keep state updates synchronized with visuals.